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# MARMARA UNIVERSITY
# FACULTY OF ENGINEERING

## EE4002
## Digital Signal Processing

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## Lab 5

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# Introduction

In previous laboratory sessions, we learned how to generate digital signals using STM32 microcontrollers on STM32CubeIDE. Building on that foundation, the goal of this lab is to take a step further by applying signal processing techniques — specifically, FIR (Finite Impulse Response) filters — to signals generated by the STM32.

As you know, a digital square wave contains not only its fundamental frequency but also many higher-order harmonics. When analyzing a square wave in the frequency domain, we observe that it consists of odd harmonics (1st, 3rd, 5th, etc.). In practical applications, we may want to isolate a specific harmonic and suppress all others. This is the purpose of the FIR band-pass filter implemented in this experiment: to extract the **1st harmonic** from a square wave and eliminate the other frequency components.

To achieve this, a square wave of 25 Hz will be generated using the STM32 DAC output, and a band-pass FIR filter will be designed and implemented in C to isolate the 50 Hz component. TIM2 will be used to trigger the DAC using DMA, allowing precise waveform generation.

You will visualize both the raw square wave and the filtered sine wave output using the Analog Discovery 2 with the WaveForms software, and you will interpret their characteristics.

In this lab you will:

- Generate a 25 Hz square wave on the STM32F446RE DAC.
- Design and implement a FIR band-pass filter in C around the 1st harmonic (50 Hz).
- Use TIM2 to trigger the DAC via DMA.
- Observe the raw and filtered waveforms on Analog Discovery 2.

You will capture each waveform, include it in your report, and comment on what you observe.

# Components

- STM32F446RE / STM32F407VG Nucleo board (DAC1 on PA4)
- USB cable for power and programming
- Analog Discovery 2 + WaveForms software
- STM32CubeIDE
- (Optional) Breadboard and wires to route PA4 to your scope

# Pre-Lab

Before the lab session:

1. Simulate a 25 Hz square wave in MATLAB or Python, sampled at 1 kHz (100 samples), and plot its frequency spectrum.
2. Design a FIR band-pass filter (approx. order 100, Hamming window) that isolates the 1st harmonic.
3. Verify via simulation that your filter suppresses higher harmonics and extracts the sinusoidal component.

**Why Pre-Lab Matters:** This pre-lab simulation ensures that the FIR filter parameters you select are appropriate before hardware implementation. Since filter behavior can be counterintuitive, it is important to validate design choices in MATLAB/Python first. This will make your waveform observations in the lab familiar and prevent confusion. Think of the pre-lab as a "virtual rehearsal" of your STM32-based experiment.

# Procedure

## 1. Raw Square Wave Generation and Visualization

- Write STM32 code to output a **25 Hz square wave** via DAC1 (PA4) with a **1 V peak-to-peak amplitude**.
- Configure TIM2 to trigger DAC updates at 1 kHz using DMA.
- Flash the code to your Nucleo board and connect PA4 to the Analog Discovery 2.
- In WaveForms, adjust timebase to show at least three cycles of the signal and set voltage scale to 0–1 V.
- Observe the square wave and note:
  - Peak-to-peak amplitude (measured vs expected).
  - Duty cycle and symmetry.
  - Rise/fall time and any noise or overshoot.
- **Important:** All screenshots must be taken with the full WaveForms window visible. The voltage scale (`V/div`) and time base (`Time/div`) must be clearly readable.

### 2. First Harmonic Extraction via FIR Filter

- Implement the FIR band-pass filter in C to extract the 50 Hz harmonic (40–60 Hz passband).
- Apply this filter in real time to the square wave samples and output the filtered signal via DAC.
- Observe the output on WaveForms — you should now see a smooth sinusoid.
- Evaluate:
  - Accuracy of the 50 Hz sine amplitude.
  - Degree of suppression of higher harmonics.
  - Residual distortion or ripple.
- **Reminder:** The same screenshot requirements apply here. Ensure that `V/div` and `Time/div` values are visible.

## Visual Analysis

For each captured figure, briefly comment on its properties:

- **Figure 1 (Square Wave):** Comment on the amplitude, duty cycle, rise/fall time, and waveform symmetry.
- **Figure 2 (Filtered 1st Harmonic):** Discuss how clean the sinusoid looks, and whether other harmonics are still visible.

## Discussion

Answer the following questions in your report:

1. Why do we use a timer (like TIM2) to control the update speed of the DAC?
2. What happens to the shape of the filtered signal if we increase the number of FIR filter coefficients?
3. How well did your filter remove the extra unwanted frequencies from the square wave?
4. Why is it important to test your filter in MATLAB or Python before running it on STM32?

## Lab Questions

Answer the following:

1. What is the purpose of applying a window (like the Hamming window) to a FIR filter?
2. How can you change the frequency of the square wave generated by the STM32?
3. What is the main trade-off when designing a FIR filter with more taps (longer filter)?

## Deliverables

Submit a single PDF report including:

- Waveform captures (Figures 1 and 2) with appropriate captions and commentary.
  **Note:** All screenshots must include the entire WaveForms window, with clearly visible voltage and time scales.
- **You do not need to paste the full `main.c` code in the report.** Only relevant function snippets and filter-related code (with comments) may be included if needed for explanation.
- Written answers to the Discussion and Lab Questions sections.